



MIXED REVISION

CHAPTERS 1 • 2 • 3

Multiple choice

- 1 Which of the following overall percentage decreases is equivalent to a percentage decrease of 5% followed by another decrease of 3%?
A 7.85% B 8% C 15% D 92% E 92.15%
- 2 An expression which simplifies to $5x - 4y$ is:
A $2x + 3y - 3x + y$ B $3x - y + 2x - 3y$ C $6x - y - x - 5y$
D $4x + 2y - 6y - x$ E $3x + y - 2x - 5y$
- 3 Adding $\begin{bmatrix} 13 & 17 \\ 21 & 19 \\ 6 & 11 \end{bmatrix}$ to $\begin{bmatrix} 15 & 27 \\ 14 & 18 \\ 29 & 33 \end{bmatrix}$ gives a sum of:
A $\begin{bmatrix} 28 & 54 \\ 35 & 37 \\ 35 & 44 \end{bmatrix}$ B $\begin{bmatrix} 28 & 44 \\ 35 & 37 \\ 35 & 44 \end{bmatrix}$ C $\begin{bmatrix} 30 & 42 \\ 40 & 32 \\ 17 & 62 \end{bmatrix}$
D $\begin{bmatrix} 54 & 28 \\ 37 & 35 \\ 44 & 37 \end{bmatrix}$ E $\begin{bmatrix} 2 & 10 \\ 7 & 1 \\ 23 & 22 \end{bmatrix}$
- 4 Nick's salary rose to \$89 950 when increased with inflation. If inflation was 3.9%, what was Nick's salary before the increase? Answer correct to the nearest dollar.
A \$86 442 B \$86 574 C \$89 946 D \$89 954 E \$93 458
- 5 The solution to the equation $\frac{2x-3}{3} = 7$ is:
A $x = 3.5$ B $x = 9$ C $x = 12$ D $x = 15$ E $x = 24$
- 6 Find the correct matrix product $\mathbf{AB}$ if $\mathbf{A} = \begin{bmatrix} 5 & 4 & 9 \\ 3 & 8 & 2 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 7 & 6 \\ 8 & 5 \\ 9 & 1 \end{bmatrix}$.
A $\begin{bmatrix} 42 & 29 \\ 37 & 25 \end{bmatrix}$ B $\begin{bmatrix} 35 & 18 \\ 32 & 40 \\ 81 & 2 \end{bmatrix}$ C $\begin{bmatrix} 148 & 103 \\ 59 & 60 \end{bmatrix}$
D $\begin{bmatrix} 148 & 59 \\ 103 & 60 \end{bmatrix}$ E $\begin{bmatrix} 35 & 24 \\ 24 & 40 \\ 81 & 2 \end{bmatrix}$
- 7 Sam can run 100 metres in 13.4 seconds. Convert this to a speed in km/h.
A 0.5 km/h B 2.1 km/h C 7.5 km/h D 26.9 km/h E 269.7 km/h

- 8 If $5 - x = 3x + 7$, then $x =$

A $-\frac{1}{2}$ B -2 C 1 D $\frac{1}{2}$ E -1

- 9 Evaluate $5\mathbf{R} - 3\mathbf{T}$ if $\mathbf{R} = \begin{bmatrix} 7 & 3 \\ 6 & 8 \end{bmatrix}$ and $\mathbf{T} = \begin{bmatrix} 10 & 6 \\ 7 & 9 \end{bmatrix}$.

A $\begin{bmatrix} -3 & -3 \\ -1 & -1 \end{bmatrix}$ B $\begin{bmatrix} 35 & 30 & 5 \\ 30 & 21 & 9 \\ 15 & 18 & -3 \\ 40 & 27 & 13 \end{bmatrix}$ C $\begin{bmatrix} 5 & -3 \\ 9 & 13 \end{bmatrix}$

D $\begin{bmatrix} -29 & -21 \\ -17 & -21 \end{bmatrix}$ E $\begin{bmatrix} -1 & -3 \\ -3 & -1 \end{bmatrix}$

Short answer questions

- Sammy the cat takes 0.4 g of vitamins in 200 mL of milk to keep his fur shiny.
 - What is this rate in mg/mL?
 - How many mg of vitamins are in 120 mL of milk?
- Expand and simplify: $a(2a - 1) - 4(3a + 2)$.
- Given that $\mathbf{A} = \begin{bmatrix} 5 & 4 \\ 11 & 8 \\ 7 & 6 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 2 & 9 & 12 \\ 10 & 5 & 6 \end{bmatrix}$, $\mathbf{C} = \begin{bmatrix} 3 & 8 \\ 1 & 0 \end{bmatrix}$ and $\mathbf{D} = \begin{bmatrix} 1 & 7 \\ 3 & 8 \\ 9 & 2 \end{bmatrix}$ answer the following questions:
 - Which pair of matrices can be added?
 - Does the matrix product $\mathbf{AC}$ exist? Explain.
 - List all of the possible matrix products formed by pairs from the given matrices.
 - State the order of each matrix product that you listed for part c.
 - Which matrix can be raised to a power? Why?
- Gemima was buying chips for her children's lunches. She found 175 g packets for \$2.49 or multipacks, with fifteen 20 g packets, for \$5.68. The multipacks were on sale for 15% off.
 - Which was the best buy?
 - What other factors could influence her decision?
- Find the value, correct to one decimal place, of A if $A = \pi r^2 + 2\pi rh$, $r = 5$ and $h = 12$.
- Evaluate:

a $\begin{bmatrix} 5 & 6 & 7 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 8 \end{bmatrix}$ b $\begin{bmatrix} 23 \\ 31 \\ 27 \end{bmatrix} - \begin{bmatrix} 18 \\ 40 \\ 26 \end{bmatrix}$

c $\begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 11 & 5 & 15 \\ 6 & 7 & 12 \end{bmatrix}$ d $\begin{bmatrix} 7 & 6 \\ 12 & 5 \end{bmatrix}^2$

Application questions

- A musical instrument store bought a saxophone for \$350. They marked it up by 45% and then added the 10% GST.
 - What was the selling price?
 - Sonya bought the saxophone at a 15% off everything sale. How much did she pay?

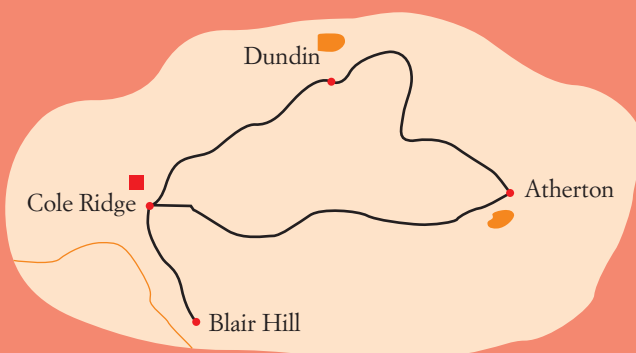
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- c Calculate the GST included in the discounted price.
 - d What was the percentage profit, correct to one decimal place, made on the sale of the saxophone?
- 2 Sahara imports jewellery from Thailand. She buys a shipment of 300 necklaces for A\$3000. The exchange rate with the Australian dollar was 36.42 baht.
 - a How much did this equate to in baht?
 - b The shipping costs were 5463 baht. How much is this equivalent to in Australian dollars?
 - c Calculate the total cost of importing the necklaces in Australian dollars.
 - d Sahara has a 220% mark-up for her profit margin. How much will she need to sell each necklace for?
- 3 The surface area of a sphere can be found using the formula $SA = 4\pi r^2$, where r is the radius of the sphere. Calculate the following, correct to one decimal place.
 - a The surface area of a sphere with a radius of 5 cm.
 - b The surface area of a sphere with a diameter of 2 m.
 - c If the surface area of a sphere is 632.5 mm^2 , calculate the radius of the sphere.
- 4 When hiring a function room there is a set fee of \$250 plus \$5 per person attending.
 - a Write an equation representing the cost (C) of hiring the function room for n people.
 - b What will it cost for a function which has 100 people attending?
 - c If the cost of a function was \$1850, how many people attended the function?
 - d The owners of the function room decide to add a cleaning fee of \$50. What would be the new equation representing the cost (C) of hiring the function room?
- 5 Helen, Katie, Grant and Robert play a dice game every night to determine who has to wash the dishes. They take it in turns rolling a pair of dice. Upon a roll of the dice, a player's score is determined by adding the sum of both dice and then bonus points if eligible.
 Bonus points are awarded for:
 - a double: + 5 bonus points
 - the sum of the 2 dice being greater than 8: + 4 bonus points.
 So if Katie rolls a pair of 3s, her score will be $(3 + 3) + 5 = 11$.
 Each player gets 5 turns.
 The results from last night are shown below:

	Sum of dice for all 5 rolls	Doubles	Number of rolls where sum > 8
Helen	29	1	1
Katie	30	0	1
Grant	23	1	2
Robert	27	2	0

- a Display this score data in a 4×3 matrix called S .
- b Grant set up a column matrix $P = \begin{bmatrix} 1 \\ 5 \\ 4 \end{bmatrix}$ which can be multiplied by S to find each player's overall score.
 - i Why is the first element a 1?
 - ii Find the matrix product SP .
 - iii The player with the lowest score for the game has to wash up. Who had to wash up last night?

6 The map below shows the roads between four properties.



We can see that we can travel directly:

- from Atherton to Dundin or Cole Ridge
- from Blair Hill to Cole Ridge
- from Cole Ridge to Atherton, Blair Hill or Dundin
- from Dundin to Cole Ridge or Atherton.

a Copy and complete the table below showing the number of direct one-step paths from each property to the others.

	Atherton	Blair Hill	Cole Ridge	Dundin
Atherton				
Blair Hill				
Cole Ridge				
Dundin				

b Display the data from the table in a matrix called M .

c Find M^2 .

d How many 2 step paths are there from:

- Atherton to Blair Hill?
- Blair Hill to Dundin?
- Cole Ridge to Blair Hill?

e Without evaluating M^4 write all the 4 step paths that you can find to travel from Blair Hill to Atherton. (Use the map to work this out).